



Visual Cues Support Robust Turn-taking Prediction in Noise

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Abstract

Accurate predictive turn-taking models (PTTMs) are essential for naturalistic human-robot interaction. However, little is known about their performance in noise. This study therefore explores PTTM performance in types of noise likely to be encountered once deployed. Our analyses reveal PTTMs are highly sensitive to noise. Hold/shift accuracy drops from 84% in clean speech to just 52% in 10 dB music noise. Training with noisy data enables a multimodal PTTM, which includes visual features to better exploit visual cues, with 72% accuracy in 10 dB music noise. The multimodal PTTM outperforms the audio-only PTTM across all noise types and SNRs, highlighting its ability to exploit visual cues; however, this does not always generalise to new types of noise. Analysis also reveals that successful training relies on accurate transcription, limiting the use of ASR-derived transcriptions to clean conditions. We make code publicly available for future research.

Index Terms: multimodal interaction, turn-taking, human-robot interaction, noise

1. Introduction

Human turn-taking is a remarkable process. There is only 200 milliseconds of silence on average between speaking turns [1], yet language production takes over 600 milliseconds [2]. Listeners must therefore plan what they are about to say whilst their conversational partner is still speaking [3]. Although effortless for humans, turn-taking is highly challenging for robots. Most turn-taking algorithms in use today are based on a robot reacting to silence after a turn [4], resulting in interactions that are less fluid than those between humans [5, 6]. Predictive turn-taking models (PTTMs) have been proposed to overcome these limitations [7, 4]. PTTMs learn human-like turn-taking from large corpora of human interaction, e.g. continuously predicting whether a shift between speakers will occur [7, 8, 9, 5], akin to human turn-taking [3].

Most PTTMs use speech features [4]. Yet in human-human interaction, listeners make faster and more accurate turn-taking decisions when they can see and hear a speaker [10]. Recent work shows multimodal PTTMs, which also use gaze, head pose and facial expression, outperform audio-only PTTMs [11, 12], illustrating the known role of visual cues in turn-taking [13, 10]. This reflects the many communicative advantages of multimodality [14], which include boosting the intelligibility of speech in noise [15]. MacLeod and Summerfield found that the intelligibility gain from being able to see a speaker was equivalent to an 11 dB increase in the signal-to-noise ratio (SNR) [15]. This motivates audio-visual speech recognition (AVSR), which exploits visual cues to overcome the performance degradation of automatic speech recognition (ASR) in noise [16].

This paper asks if multimodality confers similar performance benefits to PTTMs in noisy environments. Surprisingly, PTTMs have to date only been tested on interactions free from background noise interference [4, 11]. The impact of noise on PTTM performance is therefore unknown and unquantified. This presents a major oversight as PTTMs will be deployed *in-the-wild*, where they will inevitably encounter diverse sources and levels of noise. We therefore present the first exploration of PTTM performance in background noise, asking:

1. *How is the performance of predictive turn-taking models affected by background noise?* and
2. *Does the inclusion of visual features make multimodal PTTMs robust to background noise?*

This paper explores the crucial capability of a PTTM to distinguish holds (no speaker change) from shifts (changes between speakers). In noise-free conditions, a multimodal PTTM which incorporates visual features outperforms an audio-only model (84% and 80% accuracy, Table 1). When we add artificial music, babble, and speech noise, the accuracy of both PTTMs is poor (51% and 52% in music). Including noise during training increases accuracy (61% and 72% in music). It also ‘primes’ the multimodal PTTM to better exploit visual information (10 dB effective SNR gain over the audio-only PTTM). We find this effect generalises to types of noises unseen by the model during training. The inclusion of visual cues therefore makes PTTM performance more robust to noise. We also investigate the reliance of PTTMs training on accurate transcriptions. When noisy data is re-transcribed with ASR PTTM performance collapses (50-54%). PTTMs training therefore relies heavily on accurately aligned transcripts collected from noise-free recordings, complicating the use of ASR in noisy data.

PTTMs are essential to facilitating interaction between humans and robots, but there is little understanding of how well they could work *in-the-wild*. Part of the puzzle is understanding how they handle noise and our work therefore represents a significant step forward in PTTMs development.

2. Predictive turn-taking models

PTTMs continually predict future speaking activity, from which turn-taking predictions can be made [7, 8, 17]. For example, if speaker 0 is currently talking, but the model prediction favours speaker 1, a *shift* is predicted. PTTMs are developed on corpora of human-human interaction with the long-term eventual goal of deployment to human-robot interaction. Our work considers two-party interaction, where most work has been conducted [4].

We evaluate two PTTMs based on the state-of-the-art **voice-activity projection (VAP)** approach introduced by Ekstedt et al. [17] as it has been thoroughly explored [18, 9, 11]. VAP models predict future speaking activity in 8 bins (4 per speaker). At

time $t = 0$, bins span $[0, 0.2]$, $[0.2, 0.6]$, $[0.6, 1.2]$ and $[1.2, 2.0]$ seconds into the future. Each bin has a binary label (1=speech if $> 50\%$ frames contain speech), giving 256 possible outputs.

VAP models are neural networks consisting of stacked transformer decoder layers [17, 19]. The first model is an **audio-only VAP model** which we re-implement from [19]. The initial VAP model used mono audio and ground-truth knowledge of the current speaker [17]; i.e. it knew when each turn ended. The updated model used here takes stereo audio with 1 channel per speaker and removes the dependency on ground-truth knowledge, reflecting a real-world deployment. We also use a **multimodal VAP (MM-VAP)** model that incorporates facial action unit, facial landmark, gaze and head pose features from OpenFace [20, 11, 12]. As the initial MM-VAP model also depended on the ground-truth knowledge of the current speaker [11], we similarly extended the model to support stereo audio and remove this dependency [12]. For comparison, we also train a **video-only VAP** model, identical to VAP with the audio encoder removed [19]. Our implementations, along with full architectural details are publicly available in our repository¹.

PTTMs require labels of the start and end of each utterance for training (the **alignment**). Manual transcriptions are typically used e.g. the Switchboard telephone corpus [21, 17, 22, 5]. ASR more closely represents real-world deployments and has been shown to work well in PTTM training [23, 24].

3. Data preparation

This work uses the Candor corpus [25], an 850 hr corpus of 1657 unscripted two-party videoconferencing interactions between speakers in US English. The average session duration is 34 min. Recordings are 60 fps video, 32 kHz stereo audio with one channel per speaker. There is no controlled background noise. Recordings take place at a location of the participant's choosing e.g. home offices. We transcribe with Amazon Transcribe (EN-US model) which outputs word-level timings and preserves most disfluencies [23]. We extract gaze, head pose, facial action units, and facial landmarks with OpenFace [20].

We follow the AVSR literature by artificially simulating noisy conditions, adding noise separately to each channel at controlled SNRs [16]. **Babble noise** consists of 65,000 unique 30 speaker **babbles** which is synthesised from LRS-3 [26, 16] (46 hr train, 14 hr test). **Music noise** is sourced from MUSAN [27] (15 hr train from the *Jamendo* genre, 70 hr test). Interference from other speakers (**speech noise**) is simulated by over-laying another speaker from Candor, ensuring train/test separation.

In AVSR research, noise is typically added to isolated speech segments. However, PTTMs are trained using conversational speech which includes long pauses. A standard method of adding controlled noise to conversational speech has not been reported. We propose to add noise at a given SNR by considering signal energy during speaking times and then adding scaled noise to the complete recording. Natural variation in amplitude means this yields an average SNR over all utterances.

The **clean test audio** comprises 70 sessions withheld for evaluation. The test audio is also corrupted with -10 to +10 dB SNRs noise in 2.5 dB increments (**noisy test audio**). Shi et al. found that adding noise in training increased AVSR performance. Following their method, in each training session, with probability 0.25 we add either babble, music or speech noise at 0 dB [16], yielding the **augmented training audio**. Note that

clean and augmented training sets are the same length. We obtain both **clean** and **augmented alignments** for both training sets with Amazon ASR.

4. Model training and evaluation

First, we train with the clean audio + clean alignment, as is standard in the turn-taking literature [17, 24]. Then we train on the augmented audio + clean alignment, where noise is added but the transcription remains unchanged [16]. To test the reliance on accurate ASR, we train using the augmented audio + augmented alignment. This represents a situation where there is noisy data transcribed with ASR. Training batches consist of 20-second windows with 2 second overlap randomly sampled from training sessions. The training procedure is a 5-fold cross-validation with 10 epochs per fold on an NVidia RTX 6000 GPU. A grid search sets the batch size to 16 and the learning rate to $1e-4$. The loss function is the sum of cross-entropy (future speaking prediction) and binary cross-entropy (current speaker prediction) losses minimised with AdamW [28].

We report the hold/shift (H/S) prediction accuracy: the ability of the model to distinguish holds (no speaker change) from shifts (speaker change), a common metric in the turn-taking literature [17, 19]. First, all silences > 250 ms are identified. Those with speaker changes are shifts (83,158) and those without are holds (206,830) [17]. Then, the probability of speech in the latter half of the VAP bins is computed [17]. If speaker 0 has finished speaking and the probability of speaker 1 exceeds a binary threshold, a shift is predicted. We set the threshold to maximise balanced accuracy on validation sessions and report a 5-fold average on test sessions.

We report effective SNR gain i.e. the SNR at which MM-VAP achieves the same accuracy as VAP at 0 dB, following the AVSR literature [29]. We compare VAP and MM-VAP model performance with two-tailed independent t-tests [30].

5. Turn-taking model performance in noise

5.1. Baseline performance on clean speech

We begin our analysis by training VAP and MM-VAP models using the clean audio and clean alignment, establishing a baseline H/S prediction accuracy without added noise (clean test audio, Table 1, first two rows). The 5-fold average H/S prediction accuracy of MM-VAP model is 84%, greater than the audio-only VAP at 80% ($p < 0.001$). The accuracy and the superior performance of MM-VAP are in agreement with the PTTM literature [19, 24].

5.2. The impact of noise on turn-taking models

Figure 1 shows the average H/S accuracy on the noisy test data, where solid colours are VAP (audio-only) and hatched are MM-VAP (audio and video). We first consider the models trained on clean audio + alignment in green. The plot reveals noise has a strong impact on PTTM accuracy, most severe in speech and music where accuracy falls to 51-52% (the first two rows of Table 1). At high SNR there is a smaller impact from babble noise at 76-80% at 10 dB SNR. However, both models are extremely sensitive to music and speech interference with 51-52% accuracy at 10 dB SNR, i.e. a random hold/shift prediction.

Figure 2 shows average H/S prediction accuracy at each SNR for VAP and MM-VAP. The higher the SNR, the lower the noise energy relative to the speech signal (equal at 0 dB). Above 0 dB, the average accuracy increases as the SNR increases. Be-

¹<https://github.com/russelsa/mm-vap>

Table 1: Average hold/shift prediction accuracy (% , $\bar{x}$ =mean all SNRs) across 5-folds of cross-validation of VAP (audio, A), MM-VAP (audio+video, A+V), and video-only (V) models on the same test set at varying noise levels (2.5 dB increments, 5 dB shown for brevity). Training is conducted on the clean and augmented (aug., 25% of sessions corrupted with 0 dB noise) audio and ASR alignments.

Model	Training		Input Modality	Test set + babble @ SNR dB =						Test set + speech @ SNR dB =						Test set + music @ SNR dB =						Clean
	Audio	Alignment		-10	-5	0	5	10	$\bar{x}$	-10	-5	0	5	10	$\bar{x}$	-10	-5	0	5	10	$\bar{x}$	
VAP	clean	clean	A	61	50	58	69	76	60	51	51	51	51	52	51	51	50	51	51	52	51	80
MM-VAP	clean	clean	A+V	57	56	68	76	80	68	53	52	52	53	54	52	53	51	51	52	53	52	84
VAP	aug.	clean	A	49	50	65	74	77	63	51	59	65	66	64	61	55	58	62	66	65	61	84
MM-VAP	aug.	clean	A+V	61	62	75	79	80	71	66	69	74	74	75	72	67	68	74	76	74	72	86
VAP	aug.	aug.	A	50	50	55	55	56	53	50	50	50	49	49	50	50	50	50	50	50	50	57
MM-VAP	aug.	aug.	A+V	55	56	57	58	58	57	50	50	50	50	49	50	50	50	50	50	50	50	57
video-only	—	clean	V	66						66						66						66

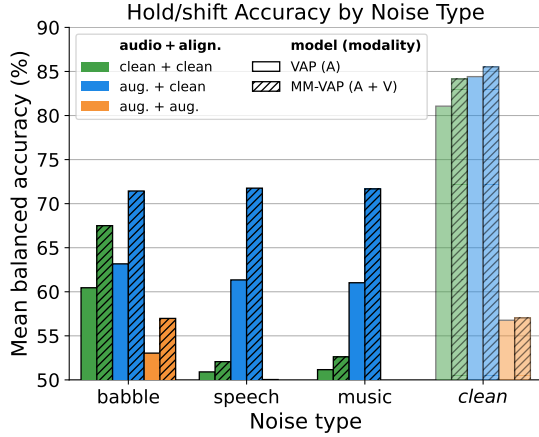


Figure 1: Average hold/shift prediction accuracy of VAP (audio-only) and MM-VAP (audio+video) models in each noise type. The average is across the -10 dB, +10 dB SNR range.

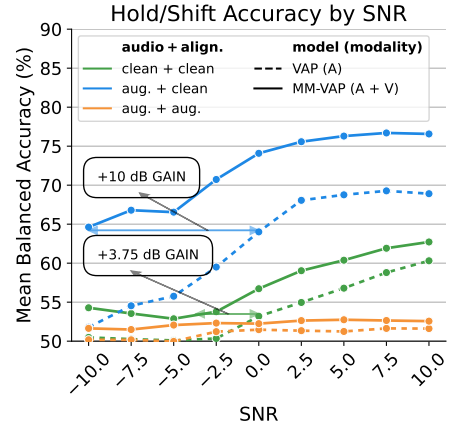


Figure 2: Average hold/shift prediction accuracy of VAP (audio-only) and MM-VAP (audio+video) models at each SNR. An average accuracy in speech, music and babble noise is shown.

low 0 dB, accuracy levels off to 50-54%. The effective SNR gain of MM-VAP over VAP is +3.75 dB. A breakdown of average accuracy by noise type reveals the superior performance of MM-VAP originates from accuracy in babble noise, as accuracy in music and speech is always 51-54% (Table 1).

5.3. Training with noisy examples

We re-train both models on the augmented training set and show H/S prediction accuracy by noise type in blue in Figure 1. There is a clear performance increase, e.g. average accuracy in music is 61% for VAP and 72% for MM-VAP, up from 51-52% ($p < 0.001$). A large gap in model accuracy has also emerged.

The blue lines in Figure 2 show performance at each SNR. At -10 dB SNR, the average accuracy of VAP is 50% whereas MM-VAP is 65% ($p < 0.01$). The effective SNR gain of MM-VAP has increased to +10 dB and the extreme sensitivity to speech and music noise is reduced. At +10 dB SNR of speech noise, the average accuracy is 64% for VAP and 75% for MM-VAP (second two rows of Table 1), up from 51% and 52% previously ($p < 0.01$). At -10 dB SNR, VAP has a 50% accuracy. On the other hand, MM-VAP backs off to the accuracy of a PTTM model trained exclusively on video features (67% vs 66% in -10 dB music noise). The only difference in input is MM-VAP receives visual features. However, it achieves superior performance in all types of noise and across the full range of SNRs, and performance is similar to a video-only model in extreme (-10 dB) noise. This demonstrates that it exploits visual information to improve accuracy in acoustic interference.

Table 2: H/S prediction accuracy (%) of MM-VAP trained with speech, babble, music, or a mixture (augmented) at 0 dB SNR.

		Test set, 0 dB SNR			
		babble	music	speech	clean
Train set	babble	71	51	52	85
	music	58	75	57	86
	speech	70	77	76	86
	augmented	75	74	74	86

5.4. Predicting a shift between speakers

We show predictions during a shift in Figure 3 (MM-VAP solid, VAP dashed) by plotting speaker 1 probability in the next 1.2-2 seconds (greater than 0.5 highlighted red). In clean audio (B), the shift between speakers is predicted ahead of time by MM-VAP. In 0 dB babble noise (C), MM-VAP predicts the shift, whereas VAP always outputs approx 0.5, demonstrating the impact of noise. MM-VAP predictions in noise (C + D) resemble predictions in clean audio (B), further demonstrating the exploitation of visual cues by MM-VAP.

5.5. Generalisation to noise unseen during training

We investigate if MM-VAP performance extends to types of noise unseen during training. We retrain three times with either 0 dB SNR speech, music or babble noise in 25% of training sessions. The balanced accuracy of H/S prediction in Table

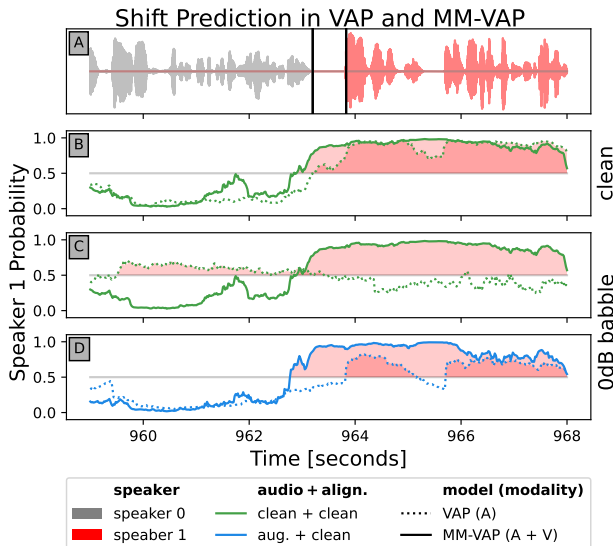


Figure 3: *Model output during a shift in the Candor corpus (unseen during training). Speaker 0 (grey): "So uhm would you feel comfortable telling me about what you do for a living?", Speaker 1 (red) "Uhm yeah well for the most part I test and tweak algorithms". VAP and MM-VAP model output is shown in clean speech (B) and 0 dB babble speech (C) and (D).*

2 shows that when the training set includes speech or babble noise, accuracy is poor in other types of noise: e.g. 71% for the babble-trained model in babble noise but only 54% in music. However, when trained on speech noise, similar accuracy is obtained in all types of noise 74 – 75% (Table 2, second-last row). This suggests training on a speech signal with an interfering speaker ‘forces’ MM-VAP to exploit visual cues, leading to better generalisation when new types of noise are encountered.

5.6. The critical role of the alignment in training

Finally, we re-train both models using the augmented alignment obtained from Amazon ASR. The orange bars in Figure 1 reveal a stark fall in accuracy to 50 – 58% and Figure 2 shows that this is consistent across all SNRs. The drop in performance to the 50% range indicates models are comparable to a random hold/shift prediction. This is due to alignment and transcription errors that ASR experiences in noise [31]. PTTM training is therefore dependent on accurate transcriptions which are either manual [17] or obtained from ASR and clean recordings [24].

6. Discussion

Our results show PTTMs do not maintain high accuracy in types of noises they are likely to encounter in-the-wild (Section 5.2). We found slight (10 dB SNR) interference from music and other speakers caused accuracy to plummet to 50%. Including noise during training dramatically increased performance (Section 5.3). This is a common technique in AVSR [16] that has not previously been considered in the turn-taking literature. Returning to our first research question, our findings therefore show that the PTTMs in this work are not robust to noise without the prior inclusion of noise in training.

The initial poor performance shows MM-VAP does not exploit visual cues to their fullest extent (Section 5.2) without appropriate training. Training on augmented data led to consider-

ably more improvement in the accuracy of MM-VAP than VAP across the SNR range. At extreme levels of noise interference (e.g. -10 dB music noise) VAP fails (50%, i.e. chance) but MM-VAP backs off to the accuracy of the video-only model (67% vs 66%, Section 5.3). This highlights the role visual cues play in achieving robust H/S accuracy in noise.

If MM-VAP is truly able to utilise visual cues to overcome noise, this should generalise to new sources of noise. However, we found that this was only true for the MM-VAP model trained on speech-noise (Section 5.5). Our working hypothesis is that mixing in random speakers over the speech signal training process ‘forces’ MM-VAP to use visual information, rather than relying on the acoustic properties of the noise. Thus, returning to our second research question, PTTMs can achieve robust performance in noise, but this is intimately related to the training process and does not inherently generalise to new noise types.

It is difficult to interpret H/S prediction accuracy numbers as no studies have determined a minimum accuracy needed to support fluid interaction in a human-robot deployment. Taking our 80-84% accuracy on clean speech as a baseline and ignoring the many complexities of a real-world deployment including hardware, real-time performance and human-machine dialogue, a robot would make a correct H/S decision 8/10 times. Both MM-VAP and VAP would only be correct 5/10 times in 10 dB SNR music interference when trained using conventional methods, significantly complicating interaction. This rises to 6.5/10 for VAP and 7.5/10 for MM-VAP when noise is included in training; close to baseline performance, illustrating the power of visual cues in achieving robust performance in noise.

A limitation of our work is we artificially add noise to speech recorded in noise-free settings. Although this is commonplace [31, 16, 29], it means we do not capture changes to verbal and non-verbal [32] communication and turn-taking [33] in noise. Another limitation is our use of videoconferencing. However, no in-person corpora match the size and quality of Candor. Finally, our mixing of noise at a constant energy level means the audio-only VAP model may be partly capitalising on the natural variations in SNR in long recordings of speech.

Training techniques for multimodal PTTMs should be further explored to ensure their ability to exploit visual cues generalises to new noise types. As AVSR can exploit visual information in datasets unseen in training [34], AVSR forms a natural starting point for future multimodal PTTMs architectures [16]. Turn-taking in noise is quite similar to the channel separation problem of speaker diarisation [35] and our findings support further exploration and the extension to the multimodal setting. There are many other turn-taking events (e.g. backchannels) which require consideration in future work.

7. Conclusion

Accurate turn-taking models are an important milestone on the path to viable human-robot interaction. We have conducted the first exploration of PTTM performance in noise, showing that PTTMs are not robust to typical noises a PTTM may encounter once deployed. Training with noise improves overall accuracy and enables multimodal PTTM to leverage visual features to overcome acoustic interference. However, this capability does not necessarily generalise to new types of noise. This issue must be solved prior to their deployment in real-world settings. Our work therefore constitutes a significant step forward in our understanding of PTTMs. Future work in PTTM architectures, training techniques and in-the-wild deployment are therefore imperative before deployment to human-robot interaction.

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